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Ecological inference and US elections

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Abstract

This thesis investigates the problem of ecological inference, which arises when we are interested in relationships between variables at the individual level but only have access to aggregate data.

Section 1 presents the problem of ecological inference, its relevance in various contexts and some of the solutions proposed. In particular, it presents the method proposed by McCartan and Kuriwaki (2025), which is the one that will be used in this thesis.

Section presents a synthetic example to illustrate the method and its performance in a controlled setting, where the true individual-level relationships are known.

Section presents an application of the method to US elections, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but we only have access to aggregate data.

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1 Introduction

Ecological inference fallacy is the problem that arises when we are interested in relationships between variables at the individual level but we only observe data at the aggregate level.

This problem is relevant in all situations in which data are reported at the aggregate level, but we are interested in relationships at the individual level.

A famous example of that problem is the one presented by Robinson (1950) for the relationship between illiteracy and race, and illiteracy and migration in the United States. In that case, the author observed that areas with higher proportion of Black population, as well as areas with higher proportion of migrants, had lower illiteracy rates. However, when looking at the individual level, Black people and migrants were more likely to be illiterate than White and non-migrant people. The explanation of this apparent paradox is that Black and migrant people were more likely to live or move to areas with lower illiteracy rates, and therefore the aggregate data masked the individual-level relationship.

We can think also at economic examples.

In any case, the most obvious case of ecological inference is in electoral behavior, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but obviously (and luckily!) we are not able to observe individual-level voting behavior, and sometimes we are not able to observe individual-level characteristics either. In these cases, we are forced to rely on aggregate data, possibly choosing the most granular level of aggregation possible. Even choosing the most granular level of aggregation possible, individual-level relationships may be highly masked. That, of course, is highly relevant for political economists, who often times deal with the relationships between demographic characteristics, electoral results

and public policies.

Kuriwaki and McCartan (2026) present a particularly convincing example of that problem, citing 1968 US presidential elections, where, in addition to the Republican and Democratic candidates Richard Nixon and Hubert Humphrey, there was also a relevant third-party candidate, George Wallace, who lead the segregationist American Independent Party. Kuriwaki and McCartan (2026) go on to show that, even at more granular levels of aggregation, such as the county level, the relationship between proportion of Black population and proportion of votes for George Wallace would still be positive considering all counties in the nation; not only that, but even restricting the analysis to some specific States, such as North Carolina, we would still observe a positive relationship: the higher the proportion of Black population, the higher the proportion of votes for George Wallace.

The key here is that, while Black people were a quite homogenous group in terms of voting behavior (generally favoring the Democratic candidate and certainly not voting for George Wallace), White people were a much more heterogeneous group, and in particular, White people with segregationist views were also more likely to live in areas with higher proportion of Black population.

Robinson (1950) came to the conclusion that ecological fallacy would constitute an inextricable problem and he concluded his paper telling that he was

... aware that this conclusion has serious consequences, and that its effect appears wholly negative because it throws serious doubt upon the validity of a number of important studies made in recent years. The purpose of this paper will have been accomplished, however, if it prevents the future computation of meaningless correlations and stimulates the study of similar problems with the use of meaningful correlations between the properties of individuals.

Econometrics in 1950, however, had considerably fewer instruments than it has now and we might think that some of these techniques could be successfully deployed in confronting ecological inference fallacy.

2 A model for ecological inference: presenting seine

Individuals are from 1 to n ; $i \in [1, N]$.

Then, we introduce the concept of geography, which is a partition of the whole population: each individual belongs to one and only one geography.

We indicate geographies by g ; $g \in [1, G]$.

We indicate geography of individual i as G_i .

Therefore, for each geography g , we have population $N_g = \sum_{i:G_i=g}$. Then, $\bar{Y}_g = \frac{\sum_{i:G_i=g} Y_i}{N_g}$.

Now, we consider the categorical variable over which we want to perform inference. That variable, too, has to be a partition of the population: each individual has to belong to one and only one of the classes encoded by the categorical variable. We indicate by d the number of possible values that the categorical variable can adopt. For example, if the partition is based on race, we might have $d = 5$ (White, Black, Hispanic, Asian, Other).

Therefore, we can indicate the categorical variable for individual i as $X_i \in \{0, 1\}^d$, where X_i is a vector of length d with all entries equal to 0 except for the entry corresponding to the value of the categorical variable for individual i , which is equal to 1.

Finally, for each geography g , we have $\bar{Y}_g = \frac{\sum_{i:G_i=g} Y_i}{N_g}$ and $\bar{X}_g = \frac{\sum_{i:G_i=g} X_i}{N_g}$.

2.1 The notation for the problem confronted by seine

Even the X are partitions.

3 A synthetic example

In this section I will show a synthetic example that helps explaining why the problem is relevant and why the solution adopted by McCartan and Kuriwaki (2025) helps us in solving it better than naive methods.

In the synthetic example we are the ones who generate the data and therefore we know the exact model of the reality. We show that, if we generate the data and then we try to estimate with different methods we obtain different results and the naive methods do not work well.

In particular, we can exploit the constitutional referendum in Italy to show a possible way in which the naive estimators could not work and therefore we can generate a dataset that encompasses the problem.

The problem is the following: we want to estimate how did voters with different levels of income vote.

If we look at data at regional level, we would be tempted to say that richest people voted more in favour of the government proposal. In fact, the three regions where the votes in favor were more than the votes in contrast (Lombardy, Veneto and Friuli-Venezia Giulia) are among the richest regions in Italy. On the contrary, the regions where No got more than 60% are among the poorest regions of Italy (Basilicata, Campania, Sicily).

However, once we look at data at municipal level, we find that, within the regions, the relationship is nearly inverse: No votes were more frequent in urban areas and other municipalities where income is relatively higher.

That divergence makes us think that if we were able to observe results at individual level, we could find even different relationships between income and voting behavior.

Therefore, we can now generate a synthetic dataset that presents a structure that resembles some stylized facts that we observe in the constitutional referendum results and other notions that we have about electoral behaviors in Italy.

Our synthetic experiment is constructed as follows.

We assume that election takes place in a Country which has three different regions: North, Centre and South.

Each Country is divided in 100 precincts, so that in the end we observe 300 precincts: 100 precincts in the North region, 100 in the Centre region, 100 in the South region.

For each electoral precinct, we observe the percentage of high-income and low-income voters and the electoral results.

Within each income category, there is high variability in their electoral behaviour.

In particular, we impose that high-income people in the North and South precincts have a high propensity of voting in favor of the reform, while in the Centre this percentage is much lower. As for low-income people, we impose that in North they have high propensity of voting in favor, while in the Centre and in the South this propensity is much lower.

At the same time, we impose that the percentage of high-income voters is highest in the North and lowest in the South.

4 An application to US data

An application to US data

5 Other Comments

Other comments adesso facciamo una prova

6 Conclusions

Conclusions

7 References

References

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